CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

*This chapter is critical because it assess whether can move from "problem statement" to a **well-reasoned system model**.*

Chapter 3 should be written in the **future tense** (what the proposed system *will* do).

3.1 Introduction

This section briefly introduces the purpose of the chapter. It explains that the chapter presents the analysis of the existing system, identifies requirements, and provides the design of the proposed system.

Example intent: to show how requirements were gathered, analyzed, and transformed into system models.

3.2 Study of the Existing System

Students describe how the current system operates (manual or computerized).

They should cover:

- How tasks are currently performed
- Who performs them
- Tools used (paper files, spreadsheets, verbal communication, etc.)
- Weaknesses and limitations of the current system

Typical weaknesses:

- Data redundancy
- Time consuming processes
- Errors due to manual handling
- Poor record keeping
- Lack of security
- Difficulty in generating reports

This section justifies the need for the new system.

3.3 Requirements Gathering Methods

Students explain **how** they collected information from users. This proves the design is based on real user needs.

Common methods:

- Interviews
- Questionnaires
- Observation
- Document analysis

For each method, students should state:

- Who was involved
- Why the method was used
- What information was obtained

3.4 System Requirements

This is one of the most important parts.

3.4.1 Functional Requirements

These describe what the system will do.

Examples:

- The system will allow user registration and login
- The system will store student/patient/customer records
- The system will generate reports
- The system will allow data update and deletion

3.4.2 Non-Functional Requirements

These describe how the system will perform.

Examples:

- Security (password protection)
- Performance (fast response time)
- Reliability
- Usability (user friendly interface)
- Portability

3.5 Feasibility Study

Students evaluate whether the system can be developed successfully.

Types of feasibility:

- Technical feasibility: availability of hardware, software, skills
- Economic feasibility: cost versus benefits
- Operational feasibility: user acceptance and ability to use the system
- Schedule feasibility: ability to complete the project on time

3.6 System Design

This section presents models and diagrams that describe how the system will be structured.

3.6.1 System Architecture

A general structure showing how components interact (e.g., client–server, web-based system).

3.6.2 Data Flow Diagrams (DFD)

Students should include:

- Context Diagram (Level 0)
- Level 1 DFD
  These show how data moves through the system.

3.6.3 Use Case Diagram

Shows interactions between users (actors) and the system.

3.6.4 Entity Relationship Diagram (ERD)

Shows database structure:

- Entities
- Attributes
- Relationships

3.6.5 Database Design (Tables)

List of tables with fields, data types, and primary keys.

3.7 Input Design

Describe how data will be entered into the system:

- Forms
- Validation rules

- Data formats

3.8 Output Design

Describe the outputs/reports the system will generate:

- Reports
- Screens
- Printouts

3.9 Process Design

Explain how data is processed in the system (algorithms, workflows, or flowcharts).

3.10 Security and Control Measures

Describe how the system will ensure:

- Authentication
- Authorization
- Data backup
- Error handling